

Chapter 9: Lower Semicontinuous Convex Functions

Convex Analysis and Monotone Operator Theory in Hilbert Spaces

BC 2e, 2017

Chapter Overview I

Main Topics:

- **Lower Semicontinuous Convex Functions ($\Gamma(\mathcal{H})$)**
 - Definition and fundamental properties
 - The lower semicontinuous envelope
- **Proper Lower Semicontinuous Convex Functions ($\Gamma_0(\mathcal{H})$)**
 - Definitions and characterization
 - Central role in this book
- **Affine Minorization**
 - Continuous affine minorants
 - Characterization through subdifferentials
- **Recession Functions**
 - Definition and properties
 - Positively homogeneous structure
- **Construction of Functions in $\Gamma_0(\mathcal{H})$**
 - Integral functions
 - Entropy and divergence examples

Lower Semicontinuity Definition

Definition 9.1: A function $f : \mathcal{H} \rightarrow [-\infty, +\infty]$ is **lower semicontinuous** (lsc) if

$$\forall x \in \mathcal{H}, \quad \liminf_{y \rightarrow x} f(y) \geq f(x)$$

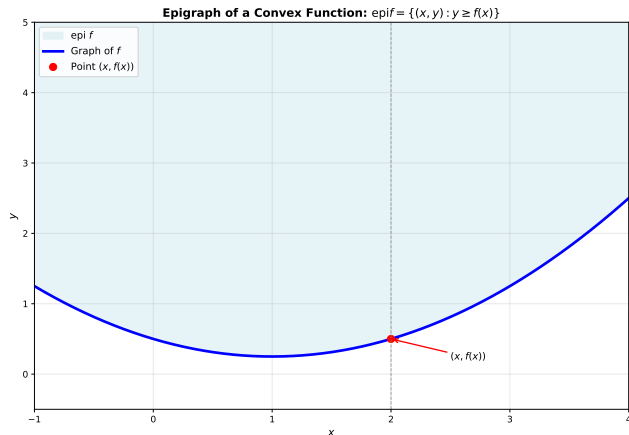
Equivalently:

- $\text{lev}_{\leq \xi} f = \{x \in \mathcal{H} : f(x) \leq \xi\}$ is closed for all $\xi \in \mathbb{R}$
- The epigraph $\text{epi} f = \{(x, y) \in \mathcal{H} \times \mathbb{R} : y \geq f(x)\}$ is closed

Key Observation: Lower semicontinuity allows jump discontinuities *upward* but not *downward*.

Notation: $\Gamma(\mathcal{H})$ denotes the class of all lsc proper convex functions on $\mathcal{H}$.

The Epigraph and Convexity



Epigraph Definition:

$$\text{epi} f = \{(x, y) \in \mathcal{H} \times \mathbb{R} : y \geq f(x)\}$$

Key Connection:

- f is convex $\iff$ $\text{epi} f$ is convex
- f is lsc $\iff$ $\text{epi} f$ is closed
- f is lsc and convex $\iff$ $\text{epi} f$ is closed and convex

Theorem 9.2: If $f \in \Gamma(\mathcal{H})$, then $\text{epi} f$ is a nonempty closed convex subset of $\mathcal{H} \times \mathbb{R}$.

Numerical Example: Convexity and LSC

Check if a function is in $\Gamma(\mathbb{R})$ — Key Properties:

For $f(x) = x^2$, verify convexity and lsc:

- ① **Convexity:** For $x_1, x_2 \in \mathbb{R}$ and $\lambda \in [0, 1]$:

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

- ② **LSC at point x :**

$$\liminf_{y \rightarrow x} f(y) \geq f(x)$$

- ③ **Closed epigraph:** The set $\{(x, y) : y \geq x^2\}$ is closed in $\mathbb{R}^2$

All these conditions are satisfied for $f(x) = x^2$, so $f \in \Gamma_0(\mathbb{R})$.

Definition and Properties of the LSC Envelope

Definition 9.7: For $f : \mathcal{H} \rightarrow [-\infty, +\infty]$, the **lower semicontinuous convex envelope** is

$$\bar{f} = \sup\{g \in \Gamma(\mathcal{H}) : g \leq f\}$$

Theorem 9.9: For $f : \mathcal{H} \rightarrow [-\infty, +\infty]$:

$$\text{epi} \bar{f} = \overline{\text{conv}(\text{epi} f)}$$

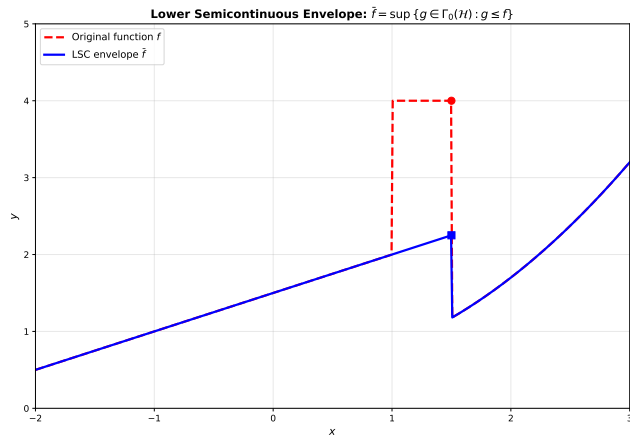
That is, the epigraph of $\bar{f}$ is the closed convex hull of the epigraph of f .

Theorem 9.10: If f is already convex, then $\bar{f} = f$.

Corollary 9.11: If f is convex and $\text{lev}_{<0} f \neq \emptyset$, then

$$\text{lev}_{<0} f \subset \text{lev}_{<0} \bar{f} \subset \text{lev}_{\leq 0} f$$

LSC Envelope Visualization



Key Properties:

- $\bar{f}$ is the *largest* lsc convex function below f
- $\bar{f} \leq f$ everywhere
- If f is already lsc and convex, then $\bar{f} = f$
- The construction “fills in” upward jumps

Practical Implication:

- Many non-convex or discontinuous functions can be “corrected” via their lsc envelope
- This provides a canonical way to regularize functions

Proper Functions

Definition 9.12: The set of **proper lower semicontinuous convex functions** from $\mathcal{H}$ to $[-\infty, +\infty]$ is denoted $\Gamma_0(\mathcal{H})$.

A function f is proper if:

- 1 f is lower semicontinuous
- 2 f is convex
- 3 $\text{dom} f \neq \emptyset$ (domain is nonempty)
- 4 $f(x) > -\infty$ for all x (never equals $-\infty$)
- 5 There exists at least one point where $f(x) < +\infty$

Remark: Nonproper lsc convex functions often have limited usefulness. Proper functions are central to this entire book.

Example 9.13: For families (e_i) and (ϕ_i) with appropriate conditions:

$$f(x) = \sum_{i \in I} \phi_i(\langle x, e_i \rangle) \in \Gamma_0(\mathcal{H})$$

Basic Properties of $\Gamma_0(\mathcal{H})$

Proposition 9.14: Let $f \in \Gamma_0(\mathcal{H})$, $x \in \mathcal{H}$, $y \in \operatorname{dom} f$. For $\alpha \in [0, 1]$:

$$\lim_{\alpha \downarrow 0} f(x_\alpha) = f(x)$$

where $x_\alpha = (1 - \alpha)x + \alpha y$.

This is a *directional lower semicontinuity* property.

Corollary 9.15: If $f \in \Gamma_0(\mathbb{R})$, then $f|_{\operatorname{dom} f}$ is continuous.

Note: This fails in general Hilbert spaces! (See Exercise 9.43 with dimension $\mathcal{H} = \mathbb{R}^2$)

Fact 9.17: For $f, g \in \Gamma_0(\mathcal{H})$:

$$\operatorname{int}(\operatorname{dom} f - \operatorname{dom} g) = \operatorname{core}(\operatorname{dom} f - \operatorname{dom} g)$$

Affine Minorization Theorem

Theorem 9.20: Let $f \in \Gamma_0(\mathcal{H})$. Then f possesses a continuous affine minorant. More precisely,

$$\exists p \in \operatorname{dom} f, \exists u \in \mathcal{H}, \quad \forall y \in \mathcal{H} : \quad \langle y - p \mid u \rangle + f(p) \leq f(y)$$

Proof Idea:

- Choose $x \in \operatorname{dom} f$ and $\xi \in (-\infty, f(x))$
- Apply the separation theorem for closed convex sets
- The separating hyperplane becomes the affine minorant

Corollary 9.21: Let $f \in \Gamma_0(\mathcal{H})$. Then f is bounded below on every nonempty bounded subset of $\mathcal{H}$.

Example 9.22: A discontinuous linear functional has no continuous affine minorant!

Supporting Affine Functions

Proposition 9.18: Let $f \in \Gamma_0(\mathcal{H})$, $(x, \xi) \in \mathcal{H} \times \mathbb{R}$, $(p, \pi) \in \mathcal{H} \times \mathbb{R}$. Then $(p, \pi) = \mathcal{P}_{\text{epi}f}(x, \xi)$ iff:

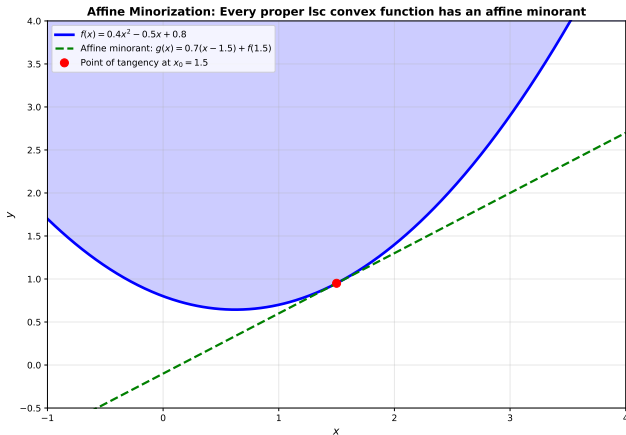
$$\max\{\xi, f(p)\} \leq \pi \quad \text{and}$$

$$\forall y \in \text{dom}f : \quad \langle y - p \mid x - p \rangle + (f(y) - f(p))(f(p) - \xi) \leq 0$$

Proposition 9.19: For $f \in \Gamma_0(\mathcal{H})$, $x \in \text{dom}f$, $\xi \in]-\infty, f(x)[$ and $(p, \pi) \in \mathcal{H} \times \mathbb{R}$:

$$(p, \pi) = \mathcal{P}_{\text{epi}f}(x, \xi) \iff \xi < f(p) = \pi \text{ and } \forall y \in \text{dom}f : \langle y - p \mid x - p \rangle \leq (f(y) - f(p))(f(p) - \xi)$$

Affine Minorization Visualization



Key Points:

- Every proper lsc convex function has supporting hyperplanes
- These hyperplanes are continuous affine functions
- They provide lower bounds (minorants)
- Fundamental for subdifferential calculus

Application - Subdifferential:

Theorem 9.23: If x is in the interior of $\text{dom } f$, then

$$\exists u \in \mathcal{H}, \exists \eta \in \mathbb{R} : \quad f(y) \geq \langle y - x \mid u \rangle + f(x)$$

This is the subdifferential condition!

Jensen's Inequality

Proposition 9.24 (Jensen's Inequality): Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, $\phi \in \Gamma_0(\mathbb{R})$, and $x : \Omega \rightarrow \mathbb{R}$ measurable with $\mu(\Omega)^{-1} \int_{\Omega} x(\omega) \mu(d\omega) \in \text{dom} \phi$. Then

$$\phi \left(\frac{1}{\mu(\Omega)} \int_{\Omega} x(\omega) \mu(d\omega) \right) \leq \frac{1}{\mu(\Omega)} \int_{\Omega} \phi(x(\omega)) \mu(d\omega)$$

Proof: Use affine minorization to find $\psi(y) = cy + d$ with $\phi(y) \geq \psi(y)$. Then integrate.

Example 9.25: If $p < q$ and X is a random variable with $\mathbb{E}[|X|^p] < \infty$:

$$\mathbb{E}[|X|^p]^{1/p} \leq \mathbb{E}[|X|^q]^{1/q}$$

This follows from Jensen with $\phi(x) = |x|^q/p^{q/p}$.

Definition of Recession Functions

Proposition 9.27: Let $f : \mathcal{H} \rightarrow [-\infty, +\infty]$ be proper and convex, $x \in \text{dom} f$, $y \in \mathcal{H}$. Set

$$\phi : \mathbb{R}_+ \rightarrow [-\infty, +\infty], \quad \alpha \mapsto \frac{f(x + \alpha y) - f(x)}{\alpha}$$

Then ϕ is increasing.

Definition 9.28: For $f : \mathcal{H} \rightarrow [-\infty, +\infty]$ proper and convex, the **recession function** at $y \in \mathcal{H}$ is

$$(\text{rec} f)(y) = \sup_{x \in \text{dom} f} \left(\frac{f(x + y) - f(x)}{1} \right)$$

Actually, by Proposition 9.27:

$$(\text{rec} f)(y) = \lim_{\alpha \rightarrow \infty} \frac{f(x + \alpha y) - f(x)}{\alpha}$$

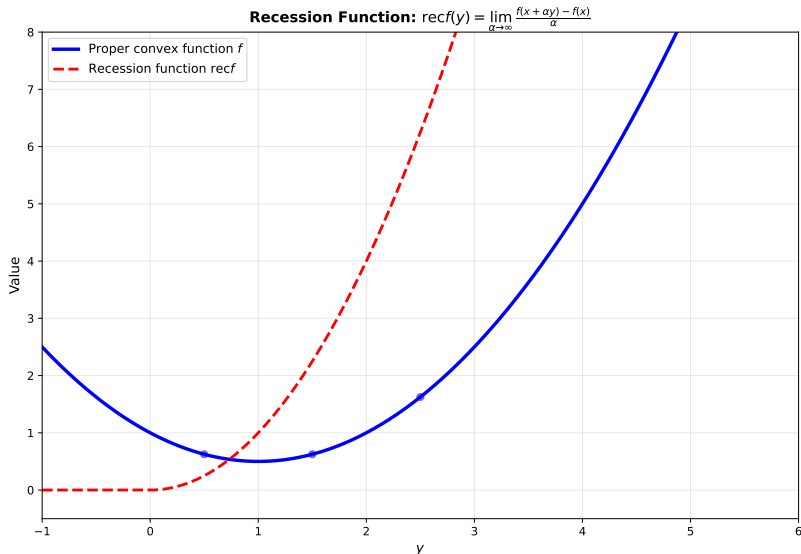
for any fixed $x \in \text{dom} f$.

Properties of Recession Functions

Proposition 9.30: Let $f \in \Gamma_0(\mathcal{H})$, $x \in \operatorname{dom} f$, $y \in \mathcal{H}$. Then:

- ① $\operatorname{rec} f \in \Gamma_0(\mathcal{H})$
- ② $(\operatorname{rec} f)(y) = \lim_{\alpha \rightarrow +\infty} \frac{f(x+\alpha y) - f(x)}{\alpha}$ (independent of x)
- ③ $(\operatorname{rec} f)(y) = \lim_{\alpha \rightarrow +\infty} f(x + \alpha y)/\alpha$
- ④ $(\operatorname{rec} f)(y) = \sup_{\alpha \in \mathbb{R}_+} \frac{f(x+\alpha y) - f(x)}{\alpha}$
- ⑤ If $\inf f(\mathcal{H}) > -\infty$, then $\operatorname{rec} f \geq 0$
- ⑥ If $g \in \Gamma_0(\mathcal{H})$ and $\operatorname{dom} f \cap \operatorname{dom} g \neq \emptyset$, then $\operatorname{rec}(f + g) = \operatorname{rec} f + \operatorname{rec} g$
- ⑦ If K is a Hilbert space, $g \in \Gamma_0(K)$, $L \in \mathcal{B}(\mathcal{H}, K)$, and $\operatorname{ran} L \cap \operatorname{dom} g \neq \emptyset$, then $\operatorname{rec}(g \circ L) = (\operatorname{rec} g) \circ L$

Recession Function Examples and Visualization



Basic Construction: The Closure-Convex Hull Extension

Proposition 9.33: Let $g : \mathcal{H} \rightarrow [-\infty, +\infty]$ be a proper convex function with $\text{dom } g$ open and g continuous on $\text{dom } g$. Set

$$f(x) = \begin{cases} g(x), & \text{if } x \in \text{dom } g \\ \lim_{y \rightarrow x} g(y), & \text{if } x \in \text{bdry dom } g \\ +\infty, & \text{if } x \notin \overline{\text{dom } g} \end{cases}$$

Then $f = \bar{g}$ and $f \in \Gamma_0(\mathcal{H})$.

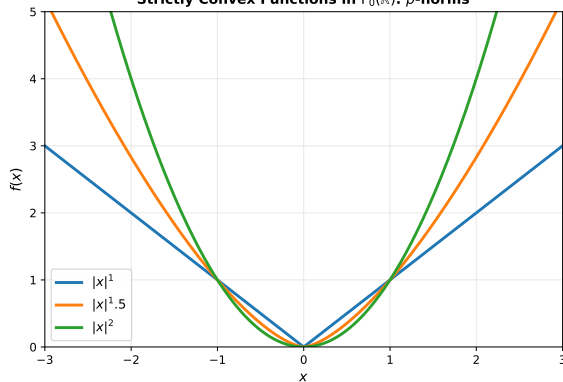
Application: This allows us to take a “partially defined” convex function and automatically extend it to a proper lsc function.

Example 9.34: For strictly convex $g : \mathbb{R} \rightarrow [-\infty, +\infty]$ on interval $[\alpha, \beta]$:

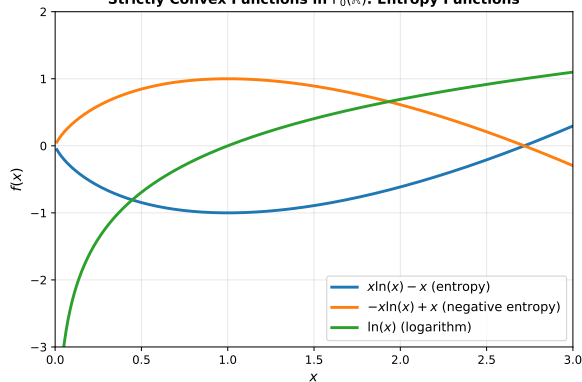
$$f(x) = \begin{cases} g(x), & \text{if } x \in [\alpha, \beta] \\ \lim_{y \uparrow \alpha} g(y), & \text{if } x = \alpha \\ \lim_{y \downarrow \beta} g(y), & \text{if } x = \beta \\ +\infty, & \text{otherwise} \end{cases} \in \Gamma_0(\mathbb{R})$$

Examples of Strictly Convex Functions in $\Gamma_0(\mathbb{R})$

Strictly Convex Functions in $\Gamma_0(\mathbb{R})$: p -norms



Strictly Convex Functions in $\Gamma_0(\mathbb{R})$: Entropy Functions



Example 9.36: The following are all strictly convex in $\Gamma_0(\mathbb{R})$:

- 1 $x \mapsto \exp(x)$
- 2 $x \mapsto |x|^p$ for $p \in [1, \infty[$
- 3 $x \mapsto 1/x^p$ (for $x > 0$, else $+\infty$) for $p \in [1, \infty[$
- 4 $x \mapsto -x^p$ (for $x \geq 0$, else $+\infty$) for $p \in]0, 1[$
- 5 Negative Fermi-Dirac entropy (for $x \in [0, 1]$)

Hölder and Young Inequalities

Proposition 9.38 (Hölder's Inequality): Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, $p \in [1, \infty[$, and $p^* = p/(p-1)$. For $x \in L^p(\Omega, \mathcal{F}, \mu)$ and $y \in L^{p^*}(\Omega, \mathcal{F}, \mu)$:

$$\int_{\Omega} |x(\omega)y(\omega)| \mu(d\omega) \leq \left(\int_{\Omega} |x(\omega)|^p \mu(d\omega) \right)^{1/p} \left(\int_{\Omega} |x(\omega)|^{p^*} \mu(d\omega) \right)^{1/p^*}$$

Example 9.39 (Hölder for Sums): For $x = (\xi_i)_{i \in I}$ and $y = (\eta_i)_{i \in I}$ in $\mathbb{R}^N$:

$$\sum_{i \in I} |\xi_i \eta_i| \leq \|x\|_p \|y\|_{p^*}$$

where $\|x\|_p = \left(\sum_{i \in I} |\xi_i|^p \right)^{1/p}$ for $p \in [1, \infty[$, and $\|x\|_{\infty} = \max_{i \in I} |\xi_i|$.

Integral Functions

Proposition 9.40: Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, $(\mathcal{H}, \langle \cdot | \cdot \rangle_{\mathcal{H}})$ a separable Hilbert space, and $\varphi \in \Gamma_0(\mathbb{R})$ with:

- ① $\mu(\Omega) < +\infty$
- ② $\varphi(0) = 0$

Set

$$f : \mathcal{H} \rightarrow [-\infty, +\infty], \quad x \mapsto \begin{cases} \int_{\Omega} \varphi(x(\omega)) \mu(d\omega), & \text{if } \varphi \circ x \in L^1((\Omega, \mathcal{F}, \mu); \mathbb{R}) \\ +\infty, & \text{otherwise} \end{cases}$$

Then $f \in \Gamma_0(\mathcal{H})$.

Intuition: Integral of proper lsc convex functions yields proper lsc convex functions.

Example: Boltzmann-Shannon Entropy

Example 9.41: Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Using convention $0 \ln(0) = 0$, set

$$f : L^2(\Omega, \mathcal{F}, \mathbb{P}) \rightarrow [-\infty, +\infty], \quad X \mapsto \begin{cases} \mathbb{E}[X \ln(X) - X], & \text{if } X \geq 0 \text{ a.s.} \\ +\infty, & \text{otherwise} \end{cases}$$

Then $f \in \Gamma_0(\mathcal{H})$.

Special Cases:

① **Entropy of random variable:** $\mathcal{H} = L^2(\Omega, \mathcal{F}, \mathbb{P})$

$$f : X \mapsto \mathbb{E}[X \ln(X) - X], \quad \text{if } X \geq 0 \text{ a.s.}$$

② **Discrete entropy:** $\mathcal{H} = \mathbb{R}^N$

$$f : (\xi_k)_{1 \leq k \leq N} \mapsto \sum_{k=1}^N \xi_k \ln(\xi_k) - \xi_k, \quad \text{if } \min_{1 \leq k \leq N} \xi_k \geq 0$$

Csiszár ϕ -Divergence

Example 9.46: For $\mathcal{H} = \mathbb{R}^N$, $\phi \in \Gamma_0(\mathbb{R})$, the **Csiszár ϕ -divergence** is a function in $\Gamma_0(\mathcal{H} \oplus \mathcal{H})$ that generalizes:

- Kullback-Leibler divergence
- Chi-squared divergence
- Hellinger distance
- And many other divergence measures

Example 9.48: The **Kullback-Leibler divergence**

$$D_{KL}(p||q) = \sum_i p_i \ln(p_i/q_i)$$

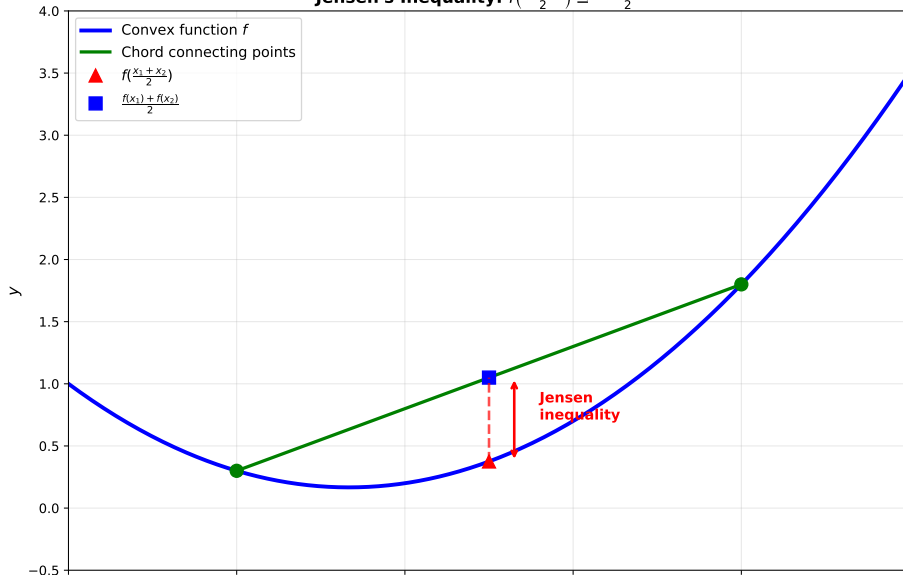
is the lsc hull of the KL divergence, and belongs to $\Gamma_0(\mathbb{R}^N \oplus \mathbb{R}^N)$.

Remark 9.47: These divergences appear naturally in:

- Information theory and statistics
- Machine learning (information bottleneck, clustering)
- Variational inference

Jensen's Inequality Visualization

$$\text{Jensen's Inequality: } f\left(\frac{x_1 + x_2}{2}\right) \leq \frac{f(x_1) + f(x_2)}{2}$$



Perspective Functions

Proposition 9.42: Let $\varphi \in \Gamma_0(\mathcal{H})$. The **perspective function** is

$$f : \mathbb{R} \times \mathcal{H} \rightarrow [-\infty, +\infty], \quad (\xi, x) \mapsto \begin{cases} \xi\varphi(x/\xi), & \text{if } \xi > 0 \\ +\infty, & \text{otherwise} \end{cases}$$

and the **recession of perspective** is

$$g : \mathbb{R} \times \mathcal{H} \rightarrow [-\infty, +\infty], \quad (\xi, x) \mapsto \begin{cases} \xi\varphi(x/\xi), & \text{if } \xi > 0 \\ (\text{rec}\varphi)(x), & \text{if } \xi = 0 \\ +\infty, & \text{if } \xi < 0 \end{cases}$$

Then $\bar{f} = g \in \Gamma_0(\mathbb{R} \oplus \mathcal{H})$.

Application: Perspective functions allow scaling with a parameter in convex optimization.

Power-Type Perspective Functions

Example 9.43: For $p \in [1, \infty[$, the function

$$g : \mathbb{R} \times \mathcal{H} \rightarrow [-\infty, +\infty], \quad (\xi, x) \mapsto \begin{cases} \|Lx - r\|^p / (\langle x | u \rangle - \rho)^{p-1}, & \text{if } \langle x | u \rangle > \rho \\ 0, & \text{if } \langle x | u \rangle = \rho \text{ and } Lx = r \\ +\infty, & \text{otherwise} \end{cases}$$

belongs to $\Gamma_0(\mathbb{R} \oplus \mathcal{H})$.

Note: This can be discontinuous at $(0, 0)$, illustrating that Γ_0 functions can be discontinuous.

Generalization (Corollary 9.44): Similar constructions work for operators $L \in \mathcal{B}(\mathcal{H}, K)$ and affine hyperplanes.

Key Takeaways I

Fundamental Concepts:

- ① **Lower Semicontinuous Functions** allow upward jumps but control behavior via:
 - Closed level sets
 - Closed epigraphs
 - The lsc envelope construction
- ② **Proper Functions** are the working class of convex analysis:
 - Never take value $-\infty$
 - Have nonempty domains
 - Form the class $\Gamma_0(\mathcal{H})$
- ③ **Affine Minorization** is crucial for subdifferential theory:
 - Every proper lsc convex function has continuous affine minorants
 - These form the subdifferential
 - Jensen's inequality follows from minorization
- ④ **Recession Functions** capture behavior at infinity:
 - Control growth rates in all directions
 - Form a proper lsc convex function themselves

Key Takeaways II

- Preserve additivity under summation

5 **Construction Methods** show the richness of $\Gamma_0(\mathcal{H})$:

- Integral functions
- Entropy and divergence examples
- Perspective functions

Key Takeaways III

Major Theorems:

- **Theorem 9.9:** Characterization of lsc envelope via closed convex hull of epigraph
- **Theorem 9.20:** Affine minorization for proper functions
- **Theorem 9.23:** Interior point characterization (subdifferential relation)
- **Proposition 9.24:** Jensen's inequality for measure spaces
- **Proposition 9.30:** Properties and structure of recession functions
- **Proposition 9.40:** Integral functions stay in $\Gamma_0(\mathcal{H})$

Connection to Later Chapters:

- Chapter 10: Subdifferentials are supported by affine minorants
- Chapter 11: Conjugate functions use proper lsc functions
- Chapters 12+: Monotone operators and fixed point theory require Γ_0 structure

Historical and Practical Context

Why This Chapter Matters:

- Lower semicontinuous convex functions are the correct framework for optimization
- Real-world problems lead naturally to non-smooth, even discontinuous, functions
- The lsc envelope allows regularization and approximation
- Affine minorization is the foundation of convex analysis
- Recession functions characterize behavior at infinity (critical for existence theory)
- Entropy and divergence functions are essential in machine learning, statistics, and information theory

Modern Applications:

- Variational methods in PDEs
- Convex optimization algorithms
- Machine learning (loss functions, regularizers)
- Information theory (entropy, divergences)
- Economic equilibrium models